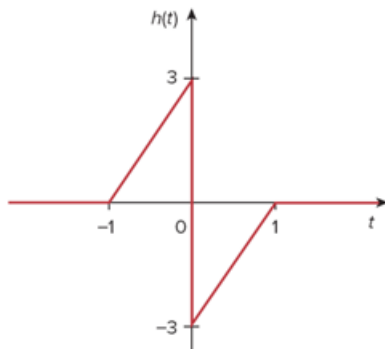


Problem

Obtain the Fourier transform of the signal shown in Fig. 18.30.

Figure 18.30



Step-by-step solution

Step 1 of 5

(Refer to Figure 18.30 in the textbook) the signal $h(t)$ is defined as,

$$h(t) = \begin{cases} 3(1+t), & -1 < t < 0 \\ 3(t-1), & 0 < t < 1 \end{cases}$$

Its first derivative is given by,

$$h'(t) = \begin{cases} 3, & -1 < t < 0 \\ -6\delta(t), & t = 0 \\ 3, & 0 < t < 1 \end{cases}$$

Step 2 of 5

The first derivative of signal $h(t)$ is shown in Figure 1.

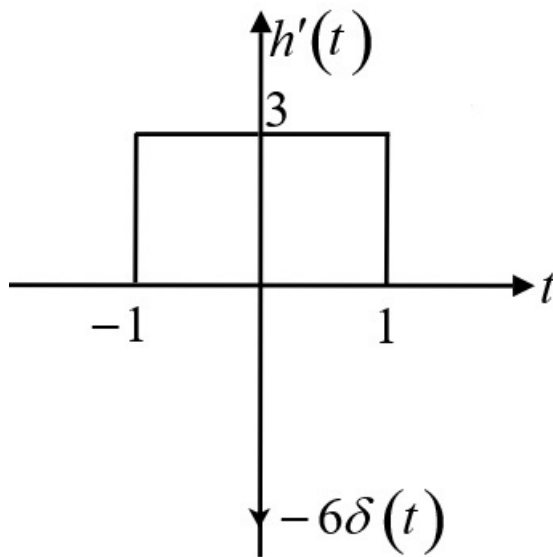


Figure 1

Step 3 of 5

Rewrite the first derivative $h'(t)$ of signal $h(t)$.

$$h'(t) = 3u(t+1) - 6\delta(t) + 3u(t-1)$$

The Fourier transform of signal $h'(t)$ is,

$$h'(t) = 3u(t+1) - 6\delta(t) + 3u(t-1)$$

$$h'(\omega) = 3 \int_{-1}^0 e^{-j\omega t} d(t) - 6 \int_{-\infty}^{\infty} \delta(t) d(t) + 3 \int_0^1 e^{-j\omega t} d(t)$$

$$h'(\omega) = \frac{3}{j\omega} (e^{j\omega} - 1) - 6 + \frac{3}{j\omega} (1 + e^{-j\omega})$$

$$h'(\omega) = \frac{3}{\omega} \left(\frac{e^{j\omega} - e^{-j\omega}}{j} \right) - 6$$

Step 4 of 5

Rewrite the above equation.

$$h'(\omega) = \frac{6}{\omega} \sin \omega - 6$$

The Fourier transform $h(t)$ using the derivative property is,

$$h(\omega) = \frac{h'(\omega)}{j\omega}$$

Step 5 of 5

Substitute $\frac{6}{\omega} \sin \omega - 6$ for $h'(\omega)$.

$$\begin{aligned} h(\omega) &= \frac{6}{j\omega^2} \sin \omega - \frac{6}{j\omega} \\ &= \frac{j6}{\omega} - \frac{j6}{\omega^2} \sin \omega \end{aligned}$$

Hence, the Fourier transform of signal $h(t)$ is $\boxed{\frac{j6}{\omega} - \frac{j6}{\omega^2} \sin \omega}$.